



Modelling Financial Markets

Seminar 13: Machine Learning in Finance — Hands-on



Daniel Traian PELE
Antoaneta Amza

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AUC: Area Under the (ROC) Curve
- ▣ CI: Confidence Interval
- ▣ CSCV: Combinatorially Symmetric Cross-Validation
- ▣ CV: Cross-Validation
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ELW: Exact Local Whittle (estimator)
- ▣ EV: Explained Variation
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FFD: Fixed-width window Fractional Differentiation
- ▣ FFN: FeedForward Network
- ▣ FN: False Negative
- ▣ GAN: Generative Adversarial Network
- ▣ GB: Gradient Boosting
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ LS: Linear SDF estimated by unregularised Least Squares
- ▣ LW: Local Whittle (estimator)
- ▣ MA: Moving Average

Acronyms used in this material (2/2)

- MA20: 20-day Moving Average
- MDA: Mean Decrease Accuracy
- MDI: Mean Decrease Impurity
- ML: Machine Learning
- OLS: Ordinary Least Squares
- PBO: Probability of Backtest Overfitting
- PSR: Probabilistic Sharpe Ratio
- PT: Pesaran–Timmermann (test of directional accuracy)
- RC: Reality Check (White's test for data snooping)
- RF: Random Forest
- ROC: Receiver Operating Characteristic (curve)
- RSI: Relative Strength Index
- SD: Standard Deviation
- SDF: Stochastic Discount Factor
- SE: Standard Error
- SHAP: SHapley Additive exPlanations
- SPA: Superior Predictive Ability (test)
- SR: Sharpe Ratio
- TN: True Negative
- TP: True Positive
- US: United States
- VIX: CBOE Volatility Index

Outline

Objectives and Setup

Part A: Pen-and-Paper Problems

Part B: Python Problems on Crypto Data

Part C: Open Problem and AI Critique

Assignment and Further Study

Key Definitions (1/4): Data, Labels, Validation

Features and labels

- **FFD**: $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$, weights truncated when $|w_k| < \tau$
- **ADF (Dickey & Fuller (1979))**: H_0 : unit root; reject if the statistic is below the 5% critical value (a rejection is not proof of stationarity: A2)
- **EWMA**: standard deviation of daily log returns with exponentially decaying weights (span 50)
- **Triple barrier**: label $+1/-1$ if the upper/lower barrier (pt , $s/\text{multiples of } \sigma_{t_0} \sqrt{h}$) is hit first; else the sign at $t_0 + h$

Validation

- **K-fold CV**: K blocks; fit on $K - 1$, test on 1, repeat
- **Leakage**: training data contain information that is also in the test data
- **Purging / embargo**: drop training observations whose label span overlaps the test block, and a few just after it

Key Definitions (2/4): Models and Classification Metrics

Models and signals

- ▣ **Logit-L1**: logistic regression with an L_1 (LASSO) penalty
- ▣ **RF**: average of decorrelated trees; **GB**: small trees added sequentially
- ▣ **MA crossover**: long when the fast MA is above the slow one

Classification metrics

- ▣ Accuracy = $\frac{TP+TN}{n}$; precision = $\frac{TP}{TP+FP}$; recall = $\frac{TP}{TP+FN}$; $F_1 = \frac{2PR}{P+R}$ (TP, FP, TN, FN: true/false positives/negatives)
- ▣ **ROC AUC**: $P(\hat{s}_+ > \hat{s}_-) + \frac{1}{2}P(\hat{s}_+ = \hat{s}_-)$ for a random positive and a random negative; 0.5 = no skill
- ▣ **Log loss**: $\ell_t = -[y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$, the loss of a probability forecast



Key Definitions (3/4): Inference and Backtests

Uncertainty

- ▣ **95% CI**: estimate ± 1.96 SE
- ▣ **Block bootstrap**: resample blocks of L consecutive observations to keep serial dependence (Künsch (1989))
- ▣ **HAC**: Newey–West variance (Newey & West (1987)), valid under serial correlation
- ▣ **DM**: Diebold–Mariano test of equal predictive accuracy: t -test on the loss differential $d_t = \ell_t^A - \ell_t^B$ with HAC SE (Diebold & Mariano (1995))

Backtests

- ▣ **SR**: $\bar{r}/\hat{\sigma}$ of daily log returns; annualised $\times \sqrt{365}$ for crypto
- ▣ **PSR / DSR** (Bailey & López de Prado (2012); Bailey & López de Prado (2014)): use skewness γ_3 , kurtosis γ_4 (Normal distribution: 3)
- ▣ N_{eff} : effective number of independent trials
- ▣ **PBO**: estimated by CSCV (Bailey, Borwein, López de Prado & Zhu (2017))

How This Seminar Works

- ▣ Two kinds of problems, in every part
 - ▶ **[Solved]**: worked models, with full solutions in this deck and in the notebook
 - ▶ **[Proposed]**: you solve them; each follows an analogous solved problem
- ▣ Part A: pen and paper (derivations, small numerical cases)
- ▣ Part B: Python on Bitcoin, verification *and* inference: every number with its uncertainty
 - ▶ each exercise ends with interpretation questions: what would change the conclusion?
- ▣ Part C: open questions that can seed your team project (C1, C3)
 - ▶ and an AI answer to critique (C2): find the leaks, correct them, log them in `AI_ERRORS.md`
- ▣ Rules: no look-ahead, purged validation, a naive benchmark, and a log of every trial ([Harvey, Liu & Zhu \(2016\)](#))

Key Definitions (4/4): Memory and Predictive Ability

Memory

- $I(\delta)$: $(1 - B)^\delta X_t$ is $I(0)$ (stationary, spectral density finite and positive at frequency 0); X_t itself is stationary only for $\delta < \frac{1}{2}$
- **LW** / **ELW**: local Whittle (Robinson (1995)) and exact local Whittle (Shimotsu & Phillips (2005)) estimators of δ from the first m Fourier frequencies; $SE = 1/(2\sqrt{m})$

Predictive ability

- **PT**: Pesaran–Timmermann test of sign independence (Pesaran & Timmermann (1992))
- **CW**: Clark–West test for nested models (Clark & West (2007)); **GW**: Giacomini–White test of conditional predictive ability (Giacomini & White (2006))
- **RC** / **SPA** / **StepM**: bootstrap tests of the best of many rules (White (2000); Hansen (2005); Romano & Wolf (2005))

Problem Overview

	[Solved]	[Proposed]
A	A1 FFD weight sums; A3 barrier hitting probabilities; A5 SR variance, PSR, DSR; A7 cost of purging; A9 AUC and its SE	A2 ADF over-rejection; A4 barriers with drift; A6 DSR and the Gumbel constant; A8 purging; A10 DeLong variance
B	B1 BTC memory; B3 triple barrier; B5 purged CV; B7 models and AUC CI; B9 meta-labelling; B10 DSR and effective N ; B12 PT and CW	B2 ETH memory; B4 asymmetric barriers; B6 find the leak; B8 Diebold–Mariano; B11 PBO; B13 SPA vs. DSR; B14 double selection
C		C1 foundation model vs. boosting; C2 critique an AI answer; C3 no-arbitrage SDF on decile portfolios

- Each proposed problem names its model: the analogous solved problem

Setup: Python Environment and Data

Environment

- Google Colab (recommended) or local Python ≥ 3.10
- `numpy`, `pandas`, `scipy`, `statsmodels`, `scikit-learn`, `shap`
-  **Open in Colab:**
`chapter13_seminar_notebook.ipynb`

```
import pandas as pd
url = 'https://raw.githubusercontent.com/danpele/MFM/main/data/market/BTC-USD.CC.csv'
btc = pd.read_csv(url, parse_dates=['date']).set_index('date')['close'].loc['2014-09-17':]
```

Data

- **Bitcoin in dollars (BTC-USD)**, daily, 2014–2026 (4 385 obs.)
- **Ether (ETH-USD)**, daily, 2015–2026 (4 061 obs.)
- Files in the course repository: `data/market`

A1 [Solved]: FFD Weights, Their Sums and the Window Length

Task

- **Context:** fractional differentiation of order d with weights $w_0 = 1$, $w_k = -w_{k-1} \frac{d-k+1}{k}$, truncated at the first $|w_k| < \tau$ (Hosking (1981); López de Prado (2018)); no data are needed
- **Tasks:**
 - ▶ (a) Show that $w_k = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)}$ and that $w_k \sim k^{-(1+d)}/\Gamma(-d)$ for large k .
 - ▶ (b) Show that $\sum_{k=0}^K w_k = \frac{\Gamma(K+1-d)}{\Gamma(1-d)\Gamma(K+1)} \sim K^{-d}/\Gamma(1-d)$.
 - ▶ (c) For $d = 0.25$ and $\tau = 10^{-4}$ compute the window K and $\sum_{k \leq K} w_k$.
 - ▶ (d) Decide whether the FFD series of an $I(1)$ log price is stationary.
- **Report:** the two formulas, K , the weight sum and the decision in (d)

Interpretation

- Why do the lecture and the literature call the FFD series “stationary”, and what is actually stationary in it?

A1: Solution (1/2)

Solution

- (a) w_k is the coefficient of B^k in $(1 - B)^d = \sum_k \binom{d}{k} (-B)^k$, and $(-1)^k \binom{d}{k} = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)}$
 - Stirling: $\Gamma(k-d)/\Gamma(k+1) \sim k^{-d-1}$, so $w_k \sim k^{-(1+d)}/\Gamma(-d)$: a hyperbolic, not exponential, decay
- (b) The partial sums of the coefficients of a series $f(B)$ are the coefficients of $f(B)/(1 - B)$; here $(1 - B)^d/(1 - B) = (1 - B)^{d-1}$
 - so $\sum_{k \leq K} w_k = \frac{\Gamma(K+1-d)}{\Gamma(1-d)\Gamma(K+1)} \sim K^{-d}/\Gamma(1-d)$, which tends to 0 only as $K \rightarrow \infty$

A1: Solution (2/2)

Solution

- (c) Set $|w_K| = \tau$: $K \approx (\tau |\Gamma(-d)|)^{-1/(1+d)} = (10^{-4} \cdot 4.902)^{-0.8} = 444$ lags (the recursion also stops at 444)
- Weight sum: $K^{-d}/\Gamma(1-d) = 444^{-0.25}/1.2254 = 0.2177/1.2254 = 0.178$
- (d) Decompose: $\tilde{X}_t = (\sum_{k \leq K} w_k) X_t - \sum_{k=1}^K w_k (X_t - X_{t-k})$
 - ▶ if X_t is $I(1)$, every $X_t - X_{t-k}$ is $I(0)$, so the second term is stationary
 - ▶ the first term is $0.178 X_t$: a scaled random walk, hence $\tilde{X}_t$ is still $I(1)$
- Reading: only the combination $X_t - X_{t-k}$ is stationary; the “memory” kept by FFD is a small multiple of the price level itself

A2 [Proposed]: Window Length, Weight Sums and ADF Over-Rejection

Task

- **Context:** the formulas of A1 with $\tau = 10^{-4}$; a Monte Carlo on pure random walks $X_t = \sum_{s \leq t} \varepsilon_s$, $\varepsilon_s \sim N(0, 1)$ i.i.d., seed 42
- **Tasks:**
 - ▶ (a) For $d = 0.3$ and $d = 0.5$ compute $K(\tau)$ and $\sum_{k \leq K} w_k$ with the closed forms of A1.
 - ▶ (b) Simulate 500 random walks, apply FFD with each d and keep $T = 2,500$ FFD values per path.
 - ▶ (c) Run the ADF test (constant, 1 lag) on each FFD path and record how often it rejects at 5%.
- **Report:** a table with d , K , the weight sum and the rejection frequency
- **Model:** A1

Interpretation

- The nominal significance level is 5%: why does the ADF confuse a scaled random walk with a stationary series? Which d would make an $I(1)$ price stationary?

A3 [Solved]: Triple-Barrier Labels and Hitting Probabilities

Task

- **Context:** entry price $P_{t_0} = 100$, daily volatility $\sigma_{t_0} = 1\%$, vertical barrier $h = 5$ days, $pt = sl = 1$, horizontal barriers $P_{t_0} e^{\pm \sigma_{t_0} \sqrt{h}}$ (López de Prado (2018)); for (b)–(c) the log price is σW_t , a driftless Brownian motion
- **Tasks:**
 - ▶ (a) Label path 1 (100, 101.0, 100.4, 102.5, 101.2, 99.8) and path 2 (100, 99.2, 98.6, 99.5, 98.9, 99.1).
 - ▶ (b) With barriers $+a = pt \sigma \sqrt{h}$ and $-b = -sl \sigma \sqrt{h}$ and no vertical barrier, derive $P(\text{upper first})$ and $\mathbb{E}[\tau]$ by optional stopping.
 - ▶ (c) For $pt = sl = 1$ compute the probability of reaching the vertical barrier, with continuous and with daily monitoring ($h = 10$).
- **Report:** the two labels, the two formulas and the two time-out probabilities

Interpretation

- The lecture finds 59.0% time-outs on the S&P 500 and B3 finds 59.0% on Bitcoin: what explains the gap to the Brownian value?

A3: Solution (1/2)

Solution

- (a) Width $0.01\sqrt{5} = 0.02236$: upper $100e^{0.02236} = 102.26$, lower 97.79
 - ▶ path 1: day 3, $102.5 \geq 102.26 \Rightarrow$ label +1 (the later fall does not matter)
 - ▶ path 2: no horizontal touch; $\ln(99.1/100) = -0.90\%$ at day 5 $\Rightarrow$ label -1
- (b) W_t is a martingale: $0 = p a - (1 - p) b \Rightarrow p = \frac{b}{a+b} = \frac{sl}{pt+sl}$
- $\sigma^2 W_t^2 - \sigma^2 t$ is a martingale: $\sigma^2 \mathbb{E}[\tau] = p a^2 + (1 - p) b^2 = ab \Rightarrow \mathbb{E}[\tau] = ab/\sigma^2 = pt \cdot sl \cdot h \text{ days (Karatzas \& Shreve (1998))}$

A3: Solution (2/2)

Solution

- (c) Exit time of $|W_s| < 1$ on $[0, 1]$ (series solution of the heat equation):
 - ▶ $P(\tau > h) = \frac{4}{\pi} \sum_{n \geq 0} \frac{(-1)^n}{2n+1} e^{-(2n+1)^2 \pi^2 / 8} = 1.2732 \cdot 0.2912 - \dots = 0.371$
- Daily monitoring misses touches between closes: shift each barrier out by 0.5826σ (Broadie, Glasserman & Kou (1997)), i.e. $c = 1 + 0.5826/\sqrt{10} = 1.184$; replacing $\pi^2/8$ by $\pi^2/(8c^2)$ gives 0.528
- Reading: with this approximate continuity correction, discrete monitoring explains most of the gap from 37% to 59%; the rest may reflect fat tails, volatility clustering and error in the EWMA volatility (which lags turbulent periods); this comparison does not separate them

A4 [Proposed]: Asymmetric Barriers with Drift

Task

- **Context:** warm-up with $P_{t_0} = 50$, $\sigma_{t_0} = 2\%$, $h = 4$, $pt = 2$, $sl = 1$; then the log price $X_t = \mu t + \sigma W_t$ with barriers $+a = pt \sigma \sqrt{h}$ and $-b = -sl \sigma \sqrt{h}$
- **Tasks:**
 - ▶ (a) Compute both barriers and label the paths (i) 50, 51.5, 53.0, 49.5, 47.9 and (ii) 50, 52.0, 54.5, 53.0, 51.0.
 - ▶ (b) Using the scale function $s(x) = e^{-2\mu x / \sigma^2}$, derive $P(\text{upper first}) = \frac{e^{2\mu b / \sigma^2} - 1}{e^{2\mu b / \sigma^2} - e^{-2\mu a / \sigma^2}}$ and show that it tends to $\frac{b}{a+b}$ as $\mu \rightarrow 0$.
 - ▶ (c) Evaluate the probability for $\sigma = 3\%$, $\mu = 0.1\%$ per day, $h = 10$, $pt = 2$, $sl = 1$.
- **Report:** the barriers, the two labels, the derivation and the number in (c)
- **Model:** A3

Interpretation

- Formula (b) has no vertical barrier: which paths does a vertical barrier at h censor, and what benchmark would you compare with B4 for $(pt, sl) = (2, 1)$?

A5 [Solved]: The Variance of the Sharpe Ratio, PSR and DSR

Task

- **Context:** i.i.d. returns with mean μ , variance σ^2 , skewness γ_3 and kurtosis γ_4 ; a strategy with $\widehat{SR}_{ann} = 1.2$ on $T = 756$ daily returns, $\hat{\gamma}_3 = -0.4$, $\hat{\gamma}_4 = 5$ (Bailey & López de Prado (2012); Bailey & López de Prado (2014))
- **Tasks:**
 - ▶ (a) Derive $\text{Var}(\widehat{SR}) \approx \frac{1}{T} \left(1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2 \right)$ by the delta method (Lo (2002)).
 - ▶ (b) Compute $PSR(0) = \Phi \left(\widehat{SR} \sqrt{T-1} / \sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2} \right)$ with the daily SR.
 - ▶ (c) Compute SR_0 and $DSR = PSR(SR_0)$ if the strategy is the best of $N = 100$ independent trials, $V = 1/T$.
- **Report:** the variance formula, $PSR(0)$, SR_0 (daily and annualised) and the DSR

Interpretation

- Would you allocate capital to this strategy? What number of trials N would make you change your mind?

A5: Solution (1/2)

Solution

- (a) $SR = g(m_1, m_2) = m_1 / \sqrt{m_2 - m_1^2}$ with $m_j = \mathbb{E}[r^j]$; gradient $\nabla g = (\frac{1}{\sigma} + \frac{\mu^2}{\sigma^3}, -\frac{\mu}{2\sigma^3})$
- Moments of (r, r^2) : $\text{Var}(r) = \sigma^2$, $\text{Cov}(r, r^2) = \sigma^3\gamma_3 + 2\mu\sigma^2$, $\text{Var}(r^2) = \sigma^4(\gamma_4 - 1) + 4\mu\sigma^3\gamma_3 + 4\mu^2\sigma^2$
- $T \text{Var}(\widehat{SR}) \approx \nabla g^\top \Sigma \nabla g$; the terms in μ^2/σ^2 cancel and leave $1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2$ (Normal distribution: $1 + \frac{1}{2} SR^2$)

A5: Solution (2/2)

Solution

- (b) Daily $\widehat{SR} = 1.2/\sqrt{252} = 0.07559$; denominator $\sqrt{1 + 0.4 \cdot 0.07559 + 1 \cdot 0.07559^2} = 1.01782$
 - ▶ $z = 0.07559 \cdot \sqrt{755}/1.01782 = 2.041$, $PSR(0) = \Phi(2.041) = 0.979$
- (c) $\sqrt{V} = 1/\sqrt{756} = 0.03637$; $\Phi^{-1}(0.99) = 2.326$, $\Phi^{-1}(1 - \frac{1}{100e}) = 2.678$
 - ▶ $SR_0 = 0.03637 [0.4228 \cdot 2.326 + 0.5772 \cdot 2.678] = 0.0920$ per day (1.46 annualised)
 - ▶ $z = (0.07559 - 0.0920)\sqrt{755}/1.01782 = -0.444$, $DSR = \Phi(-0.444) = 0.329$
- Decision: $DSR < 0.95$; luck alone gives a best SR of 1.46 > 1.2 among 100 trials: do not allocate

A6 [Proposed]: DSR on a Shorter Sample and the Gumbel Constant

Task

- **Context:** a strategy with $\widehat{SR}_{ann} = 1.5$ on 2 years of daily data ($T = 504$), $\hat{\gamma}_3 = -0.2$, $\hat{\gamma}_4 = 6$; $N = 20$ independent trials; $Z_1, \dots, Z_N$ i.i.d. $N(0, 1)$
- **Tasks:**
 - ▶ (a) Compute $PSR(0)$, SR_0 and the DSR with $V = 1/T$, step by step as in A5.
 - ▶ (b) Show that $\mathbb{E}[\max_{n \leq N} Z_n] \approx (1 - \gamma)\Phi^{-1}(1 - \frac{1}{N}) + \gamma\Phi^{-1}(1 - \frac{1}{Ne})$ in the Gumbel limit, with location $b_N = \Phi^{-1}(1 - \frac{1}{N})$ and scale $a_N = \Phi^{-1}(1 - \frac{1}{Ne}) - b_N$.
 - ▶ (c) Check (b) by Monte Carlo for $N = 100$ (20,000 draws).
- **Report:** $PSR(0)$, SR_0 , DSR, the derivation and the Monte Carlo mean
- **Model:** A5

Interpretation

- Compared with A5 the Sharpe ratio is higher and there are fewer trials: is the evidence stronger? What drives the difference?

A7 [Solved]: How Much Does Purging Cost?

Task

- **Context:** n observations in K contiguous folds of length $L = n/K$; the label of observation i ends at $t_1(i) = i + h$; embargo of e observations after each test fold (López de Prado (2018))
- **Tasks:**
 - ▶ (a) For an interior test fold, count the training observations removed before and after the fold, and the share of non-test data lost.
 - ▶ (b) Apply the result to $n = 20$, $K = 5$, $h = 3$, $e = 1$ and the test fold $\{8, \dots, 11\}$; list the purged, embargoed and training observations.
 - ▶ (c) Explain how the choice of K trades bias against variance of the CV estimate.
- **Report:** the general formula, the three sets in (b) and the trade-off in (c)

Interpretation

- Why does shuffled K-fold overstate accuracy in this setting (lecture: 69.4% on pure noise)?

A7: Solution

Solution

- (a) Before the fold: observations with $i + h \geq$ first test index overlap it: h of them
- After the fold: observations starting before the last test label ends ($i \leq$ last test index $+h$) and at least the e embargoed ones: $\max(h, e)$
- Lost share of the $n - L$ non-test observations: $\frac{h + \max(h, e)}{n - L} = \frac{K(h + \max(h, e))}{n(K - 1)}$
- (b) Purged before: $\{5, 6, 7\}$ ($t_1 \geq 8$); removed after: $\{12, 13, 14\}$ (12 = embargo, 13–14 start before day 14, when the last test label is known)
 - training set $\{0, \dots, 4\} \cup \{15, \dots, 19\}$: 6 of 16 non-test observations lost (37.5%), as the formula says: $\frac{3+3}{16}$
- (c) Larger K : more training data per fit (less pessimistic bias), but shorter and noisier test folds and more purged boundaries; when h is large relative to L , purging dominates

A8 [Proposed]: Purging with Shorter Labels

Task

- **Context:** as in A7, with $n = 20$, $K = 5$ unshuffled folds, label end $t_1(i) = \min(i + 2, 19)$ and an embargo of 1 observation
- **Tasks:**
 - ▶ (a) For the test fold $\{12, \dots, 15\}$, list the purged observations, the embargoed ones and the training set.
 - ▶ (b) Check your count against the formula of A7.
 - ▶ (c) Use the formula to compute the share of non-test data lost for $h = 20$ and folds of $L = 50$ days ($K = 5$).
- **Report:** the three sets and the two shares
- **Model:** A7

Interpretation

- What does purging cost when labels are long and folds are short, and how would you respond?

A9 [Solved]: AUC Step by Step, with a Standard Error

Task

- **Context:** classifier scores of 3 positives (0.80, 0.55, 0.40) and 3 negatives (0.60, 0.35, 0.20); Hanley–McNeil standard error (Hanley & McNeil (1982))
- **Tasks:**
 - ▶ (a) Compute the AUC as the share of (positive, negative) pairs ranked correctly, $U/(n_1 n_0)$.
 - ▶ (b) Compute $SE^2 = \frac{A(1-A) + (n_1 - 1)(Q_1 - A^2) + (n_0 - 1)(Q_2 - A^2)}{n_1 n_0}$ with $Q_1 = \frac{A}{2-A}$, $Q_2 = \frac{2A^2}{1+A}$.
 - ▶ (c) Test $H_0: AUC = 0.5$ at 5% with $z = (A - 0.5)/SE$.
- **Report:** U , AUC, SE, z and the decision

Interpretation

- Which assumption of the SE formula fails for overlapping 5-day labels, and in which direction does it bias the test?

A9: Solution

Solution

- (a) 0.80 beats 3 negatives, 0.55 beats 2, 0.40 beats 2: $U = 7$ of 9 pairs, $A = 0.778$
- (b) $Q_1 = 0.778/1.222 = 0.636$, $Q_2 = 2 \cdot 0.605/1.778 = 0.681$
 - ▶ $SE^2 = [0.173 + 2(0.636 - 0.605) + 2(0.681 - 0.605)]/9 = 0.0429$, $SE = 0.207$
- (c) $z = (0.778 - 0.5)/0.207 = 1.34$, $p = 0.18$ (asymptotic Wald approximation, unreliable here); exact permutation test with exchangeable labels under H_0 : $P(U \geq 7) = 4/20$, two-sided $p = 0.40$; do not reject: six observations cannot separate skill from luck
- The formula assumes independent observations; overlapping labels share returns, so the effective sample is smaller and the formula understates the SE: the test over-rejects. B7 replaces it by a block bootstrap

A10 [Proposed]: AUC as a U-Statistic and the DeLong Variance

Task

- **Context:** scores of 4 positives (0.70, 0.62, 0.48, 0.45) and 4 negatives (0.66, 0.50, 0.40, 0.30); kernel $\psi(s_i, s_j) = 1\{s_i > s_j\} + \frac{1}{2}1\{s_i = s_j\}$
- **Tasks:**
 - ▶ (a) Compute U , the AUC and its Hanley–McNeil SE, as in A9.
 - ▶ (b) Show that $\text{AUC} = \frac{1}{n_1 n_0} \sum_{i,j} \psi(s_i, s_j)$ is a two-sample U-statistic; define the placement values $V_{10}(i) = \frac{1}{n_0} \sum_j \psi(s_i, s_j)$ and $V_{01}(j) = \frac{1}{n_1} \sum_i \psi(s_i, s_j)$.
 - ▶ (c) Derive $\widehat{\text{Var}}(\text{AUC}) = S_{10}/n_1 + S_{01}/n_0$ (DeLong, DeLong & Clarke-Pearson (1988)), where S_{10} , S_{01} are the sample variances of the placement values, and compute the DeLong SE.
- **Report:** U , AUC, the two standard errors and the placement values
- **Model:** A9

Interpretation

- Why do both formulas assume independent observations, and what does that mean for overlapping 5-day labels?

B1 [Solved]: Memory of the Bitcoin Log Price: ADF Rule vs. Local Whittle

Task

- **Context:** daily BTC-USD closes 2014-09-17 to 2026-09-18 (4,385 obs., data/market/BTC-USD.CC.csv), $x_t = \ln P_t$; FFD with $\tau = 10^{-4}$; $m = n^{0.65}$ Fourier frequencies
- **Tasks:**
 - ▶ (a) For $d \in \{0, 0.05, \dots, 1\}$ compute the FFD series, the ADF statistic (constant, 1 lag, Dickey & Fuller (1979)) and $\text{Corr}(\tilde{x}_t, x_t)$; find the smallest d^* for which the ADF statistic is below the 5% critical value.
 - ▶ (b) Estimate the memory with 95% CIs: exact local Whittle (Shimotsu & Phillips (2005)) for x_t and the FFD series at d^* , local Whittle (Robinson (1995)) for returns and absolute returns.
 - ▶ (c) Test $H_0 : d = 1$ for the log price with $z = (\hat{d} - 1) 2\sqrt{m}$.
- **Report:** the ADF table for $d \leq 0.25$, d^* , the four estimates with CIs and the test in (c)

Interpretation

- Is the FFD series at d^* stationary? Where does the memory of Bitcoin actually sit?

B1: Solution (1/2)

Solution

- (a) ADF statistic along d (5% critical value -2.86):

d	0	0.05	0.10	0.15	0.20	0.25
ADF	-1.21	-2.32	-2.24	-2.39	-2.85	-3.38
Corr	1.000	0.999	0.995	0.985	0.972	0.955

- The rule gives $d^* = 0.25$ with correlation 0.955; the path is not monotone in d (-2.32 , then -2.24), a first sign of fragility
- The FFD weights at $d = 0.25$ sum to 0.178, so the FFD series still contains $0.178 x_t$ (A1)

B1: Solution (2/2)

Solution

- (b) Memory estimates, $m = n^{0.65}$, 95% CI $\hat{d} \pm 1.96/(2\sqrt{m})$:
 - ▶ log price (ELW): 1.065 [1.000; 1.129]; FFD series at $d^* = 0.25$ (ELW): 0.814 [0.747; 0.880]
 - ▶ returns (LW): 0.063 [-0.002; 0.127]; absolute returns (LW): 0.353 [0.289; 0.417]
- (c) $z = (1.065 - 1)/0.033 = 1.97$, two-sided $p \approx 0.05$: at most a borderline rejection, in the direction of $d > 1$ (bubble episodes), never towards stationarity
- Reading: over the band the FFD series has $\hat{d} \approx 1 - 0.25$ (its integration order is still 1, A1) and its CI lies far above $\frac{1}{2}$: it is not stationary; the long memory of Bitcoin is in volatility ($|r_t|$), as for the S&P 500 in the lecture

B2 [Proposed]: Memory of Ether: a Trap

Task

- ▣ **Context:** daily ETH-USD closes 2015-08-07 to 2026-09-18 (4,061 obs., `data/market/ETH-USD.CC.csv`); the functions of B1
- ▣ **Tasks:**
 - ▶ (a) Repeat the ADF rule of B1 on $\ln P_t^{ETH}$ and report d^* together with the ADF statistics at $d = 0$ and $d = 0.05$.
 - ▶ (b) Redo the $d = 0$ test with a constant and a trend, and redo the rule on the sample starting on 1 January 2018.
 - ▶ (c) Estimate d by exact local Whittle (95% CI) on both samples, and by local Whittle for absolute returns.
- ▣ **Report:** the three ADF-rule answers and the three memory estimates with CIs
- ▣ **Model:** B1

Interpretation

- ▣ Which d would you use in a model, and why? What does this say about unit-root tests on an asset that grew 1,000-fold?

B3 [Solved]: Triple-Barrier Labels on Bitcoin

Task

- **Context:** daily BTC-USD closes 2014–2026; EWMA volatility of daily log returns (span 50); every day is an event; vertical barrier $h = 10$ days (López de Prado (2018))
- **Tasks:**
 - ▶ (a) Label every day with symmetric barriers $pt = sl = 1$ and $pt = sl = 2$ (units of $\sigma_{i_0} \sqrt{h}$).
 - ▶ (b) Compute the shares of upper, lower and vertical exits, the average holding period and the share of +1 labels.
 - ▶ (c) Compare with the 10-day fixed-horizon label.
- **Report:** a table by setting and a histogram of the holding periods

Interpretation

- Why are about 56% of the labels +1, and what is then the benchmark accuracy of a classifier?

B3: Solution

Solution

- (a)–(b) Exit shares and labels:

	Upper	Lower	Vertical	Days	+1
$pt = sl = 1$	22.8%	18.2%	59.0%	8.25	55.7%
$pt = sl = 2$	7.2%	4.9%	87.9%	9.58	55.6%
fixed 10 days	–	–	100%	10	55.5%

- (c) Wider barriers push most exits to the vertical barrier: the labels converge to fixed-horizon labels
- Reading: BTC rose over the sample, so +1 dominates whatever the barriers; “always up” (about 56%) is the benchmark to beat, and the 59.0% time-outs are above the 52.8% of A3 (daily monitoring); the rest may reflect fat tails, volatility clustering and volatility-estimation error, which this comparison does not separate

B4 [Proposed]: Asymmetric Barriers on Bitcoin

Task

- ▣ **Context:** the data, volatility and labelling function of B3, $h = 10$
- ▣ **Tasks:**
 - ▶ (a) Repeat B3 with $(pt, sl) = (2, 1)$ and $(1, 2)$.
 - ▶ (b) Report the exit shares, the share of +1 labels and the average holding period.
 - ▶ (c) Compare the share of upper exits among horizontal exits with the infinite-horizon value $\frac{sl}{pt+sl}$ of A3 and with a simulated finite-horizon benchmark (Gaussian daily steps, same barriers, vertical barrier at h).
- ▣ **Report:** a table with the two settings and the comparison in (c)
- ▣ **Model:** B3

Interpretation

- ▣ Which setting would a trend follower choose? How does the choice change the class balance a classifier sees?

B5 [Solved]: Purged K-Fold vs. Shuffled K-Fold

Task

- **Context:** a simulated leakage dataset (seed 42): random walk of 3,000 days, label = sign of the 20-day forward return, 5 features = 20-day means of independent noise; true accuracy 50%
- **Tasks:**
 - ▶ (a) Write `PurgedKfold(n_splits, t1, pct_embargo)` with a scikit-learn `split(X)` method.
 - ▶ (b) Fit a random forest ([Breiman \(2001\)](#); 200 trees, minimum leaf 5) and compute the out-of-fold accuracy under shuffled 5-fold, unshuffled 5-fold and purged 5-fold with a 1% embargo.
 - ▶ (c) Check your class on the A7 example.
- **Report:** the three mean accuracies with the five fold values

Interpretation

- Would the gap shrink with non-persistent features? With 1-day labels?

B5: Solution

Solution

- (b) Shuffled K-fold: **69.4%**; unshuffled: 50.0%; purged + 1% embargo: **49.7%**; truth: 50%
- (c) On the A7 example the class removes $\{5, 6, 7, 12, 13, 14\}$ for the test fold $\{8, \dots, 11\}$, as derived step by step
- Reading: shuffling places near-identical neighbours in train and test; a flexible model memorises them and “predicts” pure noise 19 points above chance
- Both ingredients are needed: overlapping labels *and* persistent features; remove either one and the leak disappears (Bergmeir, Hyndman & Koo (2018))

B6 [Proposed]: Find the Leak

Task

- **Context:** the B7 modelling frame (BTC, 5-day direction, 11 features, purged 5-fold); the notebook adds a 12th feature `ma11`, and the random forest now reaches a spectacular accuracy
- **Tasks:**
 - ▶ (a) Read how `ma11` is built and rank all features by MDI.
 - ▶ (b) Compute the correlation of `ma11` and of a trailing version with the target return.
 - ▶ (c) Rerun the purged CV with the trailing version and with the 11 base features only.
- **Report:** accuracy and AUC in the three cases, the MDI ranking and the two correlations
- **Model:** B5

Interpretation

- Why can purging not catch this kind of leak? Give a second example of look-ahead in a feature pipeline.

B7 [Solved]: Model Comparison and a CI for the AUC

Task

- **Context:** BTC 2014–2026; target $y_t = 1\{\ln P_{t+5} - \ln P_t > 0\}$; 11 features (momentum 1–120 days, volatility 20/60 and their ratio, distances to the 50- and 200-day moving averages, RSI); purged 5-fold, 1% embargo; 4,181 obs., 54.8% “up”
- **Tasks:**
 - ▶ (a) Fit Logit-L1 (Tibshirani (1996)), a random forest and gradient boosting (Friedman (2001)); report fold-mean accuracy and ROC AUC against “always up”.
 - ▶ (b) Compare MDI, MDA and mean |SHAP| (Lundberg & Lee (2017)) by Spearman rank correlation.
 - ▶ (c) For the pooled out-of-fold RF probabilities compute a 95% CI for the AUC with the DeLong SE (DeLong, DeLong & Clarke-Pearson (1988)) and with a circular block bootstrap (Künsch (1989)), $L \in \{1, 20, 60\}$, 1,000 replications.
- **Report:** the model table, the rank correlations and all four intervals

Interpretation

- Why is the bootstrap interval wider? Which block length would you defend?

B7: Solution (1/2)

Solution

- (a) Fold means:

	Logit-L1	RF	GB
Accuracy	54.8%	54.2%	52.2%
ROC AUC	0.492	0.516	0.502

- No model beats “always up” (54.8%); Logit-L1 predicts “up” on 99.6% of days
- (b) Spearman: MDI-SHAP 0.75, MDI-MDA -0.42, MDA-SHAP -0.10: in-sample and out-of-sample importances disagree

B7: Solution (2/2)

Solution

- (c) RF pooled AUC = 0.512
 - ▶ DeLong: SE 0.009, CI [0.495; 0.530], $z = 1.37$, $p = 0.17$
 - ▶ bootstrap $L = 1$: SE 0.009, CI [0.494; 0.530] (same as DeLong)
 - ▶ $L = 20$: SE 0.015, CI [0.484; 0.543], 21% of replications ≤ 0.5 ; $L = 60$: SE 0.014, CI [0.485; 0.541]
- Reading: DeLong assumes independent observations; overlapping 5-day labels and volatility clusters violate it, and the dependence-robust SE is 1.7 times larger
- Block length: at least the label horizon (5) and long enough for volatility clusters; results are stable from 20 to 60 days, so $L = 20$ is defensible

B8 [Proposed]: Diebold–Mariano on Probability Forecasts

Task

- **Context:** the pooled out-of-fold probabilities of B7 and a constant forecast equal to the training share of “up” days in each fold; log loss $\ell_t = -[y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$
- **Tasks:**
 - ▶ (a) Compute the mean log loss of the four forecasts.
 - ▶ (b) For the pairs RF vs. Logit-L1, RF vs. constant and GB vs. constant compute $d_t = \ell_t^A - \ell_t^B$ and the DM statistic $\bar{d}/SE_{HAC}(\bar{d})$ (Diebold & Mariano (1995)) with Newey–West lags 4 and 20.
 - ▶ (c) Compare with the naive i.i.d. t -statistic and decide at 5%.
- **Report:** a table of log losses and of DM statistics with p -values
- **Model:** B7

Interpretation

- Can a model with AUC above 0.5 lose to a constant in log loss? What would change the conclusion?

B9 [Solved]: Meta-Labelling

Task

- **Context:** BTC 2014–2026; primary model: side of the MA20/MA50 crossover; meta-label = 1 if the primary trade is profitable at its triple-barrier exit ($pt = sl = 1$, $h = 10$) (López de Prado (2018))
- **Tasks:**
 - ▶ (a) Build the meta-labels for every day with a primary signal.
 - ▶ (b) Fit a random forest (balanced class weights) on the features plus the side, on the first 60% of the signals; test on the rest after a 10-day gap.
 - ▶ (c) Compare precision, recall, F1 and the average return per trade before (all trades) and after (only $\hat{p} > 0.5$).
- **Report:** a table with the two rows and the number of trades

Interpretation

- Precision up, F1 down: which metric matters for a trader who pays costs per trade?

B9: Solution

Solution

- Test period 2022-02 to 2026-09, 1,660 signals; the meta-model keeps 49.6% of them

	Prec.	Recall	F1	Trades	Ret./trade
Primary only	0.475	1.000	0.644	1,660	−0.27%
+ meta-model	0.517	0.541	0.529	824	+0.19%

- Precision rises by 4 points, recall halves and F1 falls; the average return per trade turns from −0.27% to +0.19%
- Reading: with costs per trade, precision and the return per trade matter, not F1; the next step is a test of that return (block bootstrap) and a DSR over all meta-model settings tried

B10 [Solved]: Grid Search, DSR and the Effective Number of Trials

Task

- **Context:** long/cash MA crossover on the BTC log price, 15 May 2015 – 2026 ($T = 4,145$ days, the first day on which every rule has a signal); fast window $\in \{5, 10, \dots, 55\}$, slow $\in \{20, 40, \dots, 240\}$, slow $>$ fast: $N = 120$ trials (Bailey & López de Prado (2014))
- **Tasks:**
 - ▶ (a) Pick the best trial by Sharpe ratio and compute its $\text{PSR}(0)$.
 - ▶ (b) From the 120×120 correlation matrix of trial returns compute the eigenvalues, the participation ratio $(\sum \lambda_i)^2 / \sum \lambda_i^2$, the smallest number of leading eigenvalues whose cumulative sum accounts for 95% of the total and $\hat{N} = \bar{\rho} + (1 - \bar{\rho})N$.
 - ▶ (c) Compute SR_0 and the DSR for $N = 120$, $\hat{N}$ and the 95% count, with $V = 1/T$.
- **Report:** the best rule, its SR and moments, the effective- N measures and a DSR table

Interpretation

- Which N and which $V[\widehat{SR}]$ would you report to an investment committee?

B10: Solution

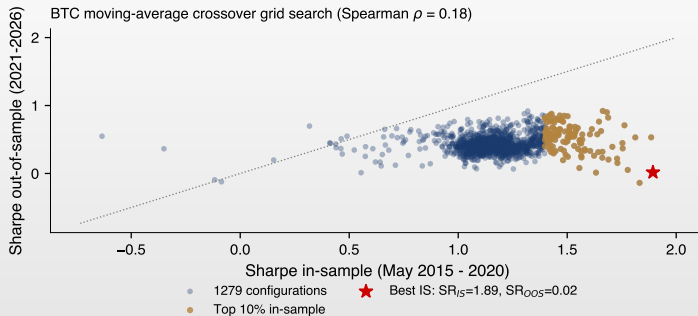
Solution

- (a) Best (5, 40): $SR_{ann} = 1.19$ (buy & hold 0.78); skewness 0.13, kurtosis 12.8; $PSR(0) = 1.000$
- (b) $\lambda_1 = 105.5$ explains 87.9% of the variance; participation ratio 1.3; 4 eigenvalues reach 95%; $\bar{\rho} = 0.875 \Rightarrow \hat{N} = 15.9$

DSR, $V = 1/T$	$N = 120$	$\hat{N} = 15.9$	$N = 4$
SR_0 (ann.)	0.77	0.53	0.31
DSR	0.922	0.987	0.998

- (c) Counting 120 correlated trials as independent over-deflates (0.922); with $\hat{N}$ the rule passes (0.987)
- Reading: the DSR tests “SR above luck”, not “above buy & hold”; report $\hat{N}$ with its method, and test against buy & hold separately (B13)

B10: Why Out-of-Sample Evidence Still Matters



- Lecture result: the best in-sample BTC crossover (SR 1.89, May 2015 – 2020) earned SR 0.02 in 2021–2026

B11 [Proposed]: Probability of Backtest Overfitting (CSCV)

Task

- **Context:** the $T \times 120$ matrix of daily returns of the B10 trials, split into $S = 16$ contiguous blocks (Bailey, Borwein, López de Prado & Zhu (2017))
- **Tasks:**
 - ▶ (a) For each of the $\binom{16}{8} = 12,870$ choices of 8 in-sample blocks, pick the trial with the best in-sample Sharpe ratio.
 - ▶ (b) Compute its relative out-of-sample rank ω and the logit $\lambda = \ln \frac{\omega}{1-\omega}$.
 - ▶ (c) Report $PBO = P(\lambda \leq 0)$, the median logit and the median in- and out-of-sample Sharpe ratio of the winner.
- **Report:** PBO, the histogram of λ and the Sharpe ratios in (c)
- **Model:** B10

Interpretation

- Is the grid overfit? How would S , or adding 1,000 random strategies to the grid, change the PBO?

B12 [Solved]: Pesaran–Timmermann and Clark–West on the BTC Direction Models

Task

- **Context:** the pooled out-of-fold probabilities of B7 ($n = 4,181$, 54.8% “up”), predicted sign $x_t = 1\{\hat{p}_t > 0.5\}$, and the constant forecast of B8 (Pesaran & Timmermann (1992); Clark & West (2007))
- **Tasks:**
 - ▶ (a) For each model compute $\hat{P}$, $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$ and the PT statistic.
 - ▶ (b) Because the 5-day labels overlap, compute the slope t -statistic of $y_t = a + b x_t + u_t$ with Newey–West errors (4 lags).
 - ▶ (c) Test Logit-L1 against the constant (nested) with the Clark–West statistic on the Brier loss, one-sided, HAC with 4 lags; compare with the plain DM test on the same loss.
- **Report:** a table with $\hat{P}$, $\hat{P}^*$, PT and the HAC t ; the CW and DM statistics with p -values

Interpretation

- Why can DM and Clark–West disagree for nested models? Which null does each test?

B12: Solution (1/2)

Solution

- (a)–(b) Directional tests:

	Pred. up	$\hat{P}$	$\hat{P}^*$	PT	t_{HAC}
Logit-L1	99.6%	0.548	0.548	0.89	0.82
RF	70.3%	0.542	0.519	3.20	2.21
GB	68.5%	0.522	0.518	0.65	0.45

- RF: its accuracy (54.2%) is below “always up” (54.8%), yet its signs are not independent of the outcome: $PT = 3.20$; the HAC correction shrinks the evidence to $t = 2.21$ ($p = 0.03$)
- Logit-L1 and GB: no significant directional information

B12: Solution (2/2)

Solution

- (c) Clark–West, Logit-L1 vs. constant: $t = -0.56$, one-sided $p = 0.71$: no evidence that the 11 features add information in the population model
- DM on the Brier loss: $t = 1.37$ (positive = logit worse), $p = 0.17$
- DM asks whether the *estimated* logit forecasts better; under the nested null its extra noise makes it lose, so DM rarely favours the larger model
- CW adds back $(\hat{\rho}_0 - \hat{\rho}_1)^2$ and asks whether the population coefficients are zero; here both agree: no significant signal
- CW is derived for nested linear models and genuine out-of-sample forecasts; on penalised-logit probabilities from pooled purged CV it is an approximate, descriptive adjustment
- Reading: a significant PT (RF) does not make a model useful: its log loss is still worse than the constant (B8)

B13 [Proposed]: SPA vs. DSR on the 120-Rule Grid

Task

- **Context:** the $T \times 120$ return matrix of B10 ($T = 4,145$); benchmarks cash ($r_0 = 0$) and buy-and-hold; stationary bootstrap (Politis & Romano (1994)) with mean block 20 days, 2,000 replications, seed 42
- **Tasks:**
 - ▶ (a) Compute the performance differentials $d_{k,t} = r_{k,t} - r_{0,t}$ for both benchmarks.
 - ▶ (b) Compute White's Reality Check p -value (White (2000)), Hansen's SPA p -value with the consistent recentring (Hansen (2005)) and the number of rules kept by the Romano–Wolf StepM at 5% (Romano & Wolf (2005)).
 - ▶ (c) Compare with the DSR of B10 under $N = 120$ and $\hat{N} = 15.9$.
- **Report:** a table with the largest studentised statistic, the two p -values and the StepM count per benchmark
- **Model:** B10 and B12; the lecture slide “SPA vs. DSR on the Bitcoin Grid”

Interpretation

- Which method handles correlated trials without choosing N ? Does the grid beat buy-and-hold?

B14 [Proposed]: Double Selection: Does the VIX Predict Bitcoin Returns?

Task

- **Context:** the B7 frame ($n = 4,181$); outcome $y_t = 5$ -day forward BTC log return in basis points; regressor $d_t = \log$ VIX close of the same calendar date, carried forward over weekends and US holidays (data/market/VIX.INDX.csv), standardised; 5 days = 5 BTC calendar days; controls: the 11 features, their squares and pairwise products ($p = 77$), standardised
- **Tasks:**
 - ▶ (a) Run a LASSO of y_t on (d_t, x_t) with the plug-in penalty $\lambda = 2c\sqrt{n}\Phi^{-1}(1 - \gamma/(2p))$, $c = 1.1$, $\gamma = 0.1/\log n$ (Belloni, Chernozhukov & Hansen (2014)); is the VIX kept?
 - ▶ (b) Run the two selection LASSOs (y_t on x_t , d_t on x_t) and report $|\hat{S}_y|$ and $|\hat{S}_d|$.
 - ▶ (c) Estimate OLS of y_t on d_t and the union of the selected controls, with HAC errors (4 lags); give $\hat{\theta}$, SE and the 95% CI.
- **Report:** the answer to (a), the two set sizes and the post-double-selection estimate
- **Model:** the lecture example “Does the VIX Predict S&P 500 Returns?” and B12

Interpretation

- Why does the single LASSO give no valid inference? What does the CI say about a crypto “fear premium”?

C1 [Proposed]: Does a Foundation Model Beat Boosting?

Task

- **Context:** the BTC 5-day direction of B7 (daily closes 2014–2026); a time-series foundation model of Chapter 14 used zero-shot; baselines of B7/B8; a team project with no single correct method
- **Tasks:**
 - ▶ (a) Choose a model and restrict the test window to dates after its pre-training cut-off, otherwise the test leaks.
 - ▶ (b) Convert the forecast quantiles of the 5-day return into $P(r_{t,t+5} > 0)$ by interpolation.
 - ▶ (c) Refit gradient boosting and the constant forecast at every forecast origin on a fixed-length rolling window, and compare the log losses on exactly the same dates with the Giacomini–White test ([Giacomini & White \(2006\)](#); HAC errors).
 - ▶ (d) Define a long/cash rule on $P(\text{up}) > 0.5$ and compute its DSR, or the SPA p -value, over all configurations you tried.
- **Report:** a Quantlet folder in the team repository (code, `Metainfo.txt`, one chart) and one slide with the conclusion and its main limitation

Interpretation

- If the foundation model wins, is the gain economically meaningful after costs, and would it survive the next five years of data?

C2 [Proposed]: Critique an AI Answer

- A student asks an AI assistant: “Predict tomorrow’s direction of Bitcoin from past returns and evaluate the model out of sample with 5-fold cross-validation”
- The assistant answers with this code (btc: daily closes, imports omitted):

```
r = np.log(btc).diff()
X = pd.DataFrame({'r1': r, 'r5': r.rolling(5).sum(), 'vol20': r.rolling(20).std(),
                  'trend10': r.rolling(10, center=True).mean()})
data = pd.concat([X, r.shift(-1).rename('fwd')], axis=1).dropna()
y = (data['fwd'] > 0).astype(int) # 1 if tomorrow is up
Xs = StandardScaler().fit_transform(data[X.columns])
cv = KFold(n_splits=5, shuffle=True, random_state=0)
model = LogisticRegression(penalty='l1', C=0.1, solver='liblinear')
auc = cross_val_score(model, Xs, y, cv=cv, scoring='roc_auc')
print(round(auc.mean(), 3))
```

- “Output: 0.697. An out-of-sample AUC of 0.70 on 4361 days is a strong, tradable signal; cross-validation is out of sample, so there is no overfitting”

C2 [Proposed]: Critique an AI Answer, Task

Task

- The answer contains **two** leaks of test information into training, a validation design that does not match deployment and **two** unsupported claims
- **Context:** the code and claim of the previous slide, run on daily BTC-USD closes 2014–2026 (data/market/BTC-USD.CC.csv)
- **Tasks**, for each error
 - ▶ (a) Say what is wrong and why.
 - ▶ (b) Say how you would detect it: a check in the code, a result from B5–B7, a rule of the lecture.
 - ▶ (c) Give the corrected code.
 - ▶ (d) Rerun the corrected code with a walk-forward split and report the AUC.
- **Report:** the five errors as in AI_ERRORS.md: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- Which days enter trend10 on day t ? Which of them is the target?
- Is an AUC of 0.70 for next-day Bitcoin direction plausible, given B6 and B7?
- Why is “cross-validation is out of sample” not enough for time series?

C3 [Proposed]: Data Specification

- Source: the value-weighted decile portfolios of the 15 univariate sorts of the [Kenneth French Data Library](#)
 - ▶ size, book-to-market, operating profitability, investment, earnings/price, cash flow/price, dividend yield, momentum, short-term reversal, long-term reversal, accruals, market beta, net share issues, variance, residual variance
- Excess returns: subtract the one-month bill rate of the library's factor file
- Legs: per characteristic a short leg (equal-weighted mean of deciles 1–5) and a long leg (mean of deciles 6–10), both entered with a positive sign (θ sets the sign): 30 factors $\tilde{F}_{t+1}$
- Sample: only months in which all 30 legs and the FF5 factors exist; no imputation
- Limitation: halves of value-weighted deciles replace the rank-weighted legs of 46 characteristics on individual stocks, so the published LS and EN values cannot be reproduced

C3 [Proposed]: A No-Arbitrage SDF on Characteristic-Sorted Deciles

Task

- ▣ **Context:** the lecture case study (Chen, Pelger & Zhu (2024)): FF3/FF5 (Fama & French (2015); Table A.VI) are reproducible with library data; the linear SDF (Section II.F) needs the stock panel, so here it is an extension, not a replication
- ▣ **Tasks:**
 - ▶ (a) Build the monthly panel, 1967 to the latest month: the 30 legs $\tilde{F}_{t+1}$ specified on the previous slide.
 - ▶ (b) On 1967–1986 compute $\hat{\theta}_{LS} = \hat{\Sigma}^{-1} \hat{\mu}$ (sample second moment and mean of $\tilde{F}_{t+1}$), $\hat{\theta}_{EN} = \arg \min_{\theta} \|\hat{\mu} - \hat{\Sigma} \theta\|_2^2 + \lambda_2 \|\theta\|_2^2 + \lambda_1 \|\theta\|_1$ (Section II.F), (λ_1, λ_2) chosen by the Sharpe ratio on 1987–1991 over a reported grid, and the FF3 and FF5 tangency weights.
 - ▶ (c) With weights fixed, report the monthly Sharpe ratio of each SDF portfolio $F_{t+1} = \hat{\theta}^\top \tilde{F}_{t+1}$ on 1992–2016, with a 95% CI (Lo (2002)).
 - ▶ (d) Estimate $\hat{\beta}_i \propto \overline{R_i^e F}$ (unconditional analogue of the paper's betas) on 1967–1986, project test returns on $\hat{\beta}$, compute EV and cross-sectional R^2 per characteristic (Section II.F); repeat (c)–(d) from 2017 on.
- ▣ **Report:** Sharpe ratios with CIs (Tables I, A.VI); EV and cross-sectional R^2 (qualitative comparison with EN, Table III); a Quantlet folder
- ▣ **Model:** A5 (Sharpe ratio inference) and B12 (predictive-ability tests)

Interpretation

- ▣ Why can the elastic net linear no-arbitrage SDF beat a forecasting network (Table I)? Does your post-2016 result survive the configurations tried?

Take-Home Assignment (Team Project)

- Choose one asset (equity index, stock, crypto, exchange rate or commodity) and deliver
 - ▶ features without look-ahead; for any fractional feature, a memory estimate (local Whittle) with its CI, chosen on training data only (B1–B2)
 - ▶ triple-barrier labels with a justified (pt, sl, h)
 - ▶ at least three models with Purged K-Fold, a Pesaran–Timmermann test, a DM or Clark–West test against a constant forecast and a block-bootstrap CI for the AUC
 - ▶ a log of every trial, the effective N , the DSR, the SPA p -value against buy-and-hold and the PBO of the final strategy
- GitHub repository with notebooks and Quantlet Metainfo.txt; a 10-minute presentation
- Log every AI error you catch in AI_ERRORS.md, as in C2: what was asked, what was wrong, how you found out, the correct version
- A negative result, honestly documented, is a good result

Further Study on Quantinar

- ▣  [Random Forests](#) — bagging, out-of-bag error, variable importance
- ▣  [Shapley Values](#) — model-agnostic local explanations
- ▣  [Machine Learning in Financial Risk](#) — ML estimation of risk measures
- ▣  [Introduction to Reinforcement Learning](#) — Q-learning, sequential decisions

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